# SMART BATTERY HEALTH MONITORING SYSTEM FOR EVs

B.Tech Mini Project Documentation

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|---|---|
| **Project Domain** | Electronics and Communication Engineering (ECE) / Electric Vehicles / IoT / Web Applica |
| **Project Type** | B.Tech Mini Project |
| **Development Platform** | Base44 AI |
| **Application Type** | Web Application |
| **Live Application** | https://arboreal-volt-pulse-track.base44.app |

Prepared for Academic Mini Project Submission

Note: The current project is a web-based prototype using simulated/demo battery data. Hardware integration with ESP32/Arduino and sensors is proposed as future scope.

# 1. Abstract

Electric Vehicles (EVs) depend heavily on batteries for their performance, range, and safety. Monitoring battery parameters such as State of Charge (SOC), State of Health (SOH), voltage, current, temperature, and charging cycles is important for maintaining battery performance and identifying abnormal conditions.

The Smart Battery Health Monitoring System for EVs is a web-based monitoring application designed to provide a simple and user-friendly interface for observing important EV battery parameters. The application provides a dashboard for monitoring battery status, analyzing battery health, viewing alerts, examining historical information, and understanding possible maintenance requirements.

The web application was developed using Base44 AI, an AI-assisted application development platform that supports creating web applications from natural-language requirements.

# 2. Problem Statement

EV batteries are complex systems whose performance changes with usage, charging cycles, temperature, current, and battery age. Without proper monitoring, abnormal conditions such as high battery temperature, low battery charge, overvoltage, undervoltage, excessive current, and battery degradation may go unnoticed.

Therefore, there is a need for a simple monitoring platform that can display battery parameters, identify abnormal conditions, provide alerts, and help users understand the overall battery condition.

# 3. Proposed Solution

The proposed system provides a web-based dashboard for monitoring and analyzing EV battery parameters. The system allows users to monitor battery status, view SOC and SOH, monitor voltage/current/temperature, track charging cycles, visualize performance, identify abnormal conditions, generate alerts, view battery history, obtain maintenance recommendations, and simulate battery sensor data.

The application can be extended in the future to receive real-time data from an ESP32/Arduino-based battery monitoring system.

# 4. Objectives

- Develop a web-based EV battery monitoring system.
- Monitor important battery parameters.
- Display battery health in an easy-to-understand format.
- Identify abnormal battery conditions.
- Provide warning and critical alerts.
- Visualize historical battery data.
- Provide basic predictive maintenance information.
- Create a platform that can later be connected to real sensors.
- Demonstrate AI-assisted application development using Base44.

# 5. Key Features

## 5.1 Dashboard

- SOC
- SOH
- Battery voltage
- Battery current
- Battery temperature
- Battery power
- Charging status
- Estimated range
- Charging cycles
- Overall battery condition

### 5.2 Battery Monitoring

- Voltage, current, temperature, SOC, SOH, power, charging/discharging status, and charging cycle count. Charts can be used to visualize parameter changes.

### 5.3 Battery Health Analysis

- Health classification: Healthy, Warning, or Critical. Recommendations are provided when abnormal values are detected.

### 5.4 Alerts and Notifications

- High/low temperature, overvoltage, undervoltage, overcurrent, low SOC, rapid discharge, poor battery health, and excessive charging cycles. Alerts can include severity, parameter, date/time, and recommended action.

### 5.5 Battery History

- Battery health trends, SOC trends, temperature trends, charging/discharging history, energy usage, and charging cycles.

### 5.6 Predictive Maintenance

- Estimated battery health, degradation information, possible battery problems, maintenance recommendations, and battery risk indication. Predictions are estimates for demonstration and are not a replacement for professional diagnostics.

### 5.7 Sensor Simulation

- Simulated voltage, current, temperature, SOC, SOH, power, and charging-cycle values allow demonstration without a physical EV battery.

## 6. Proposed Hardware Integration

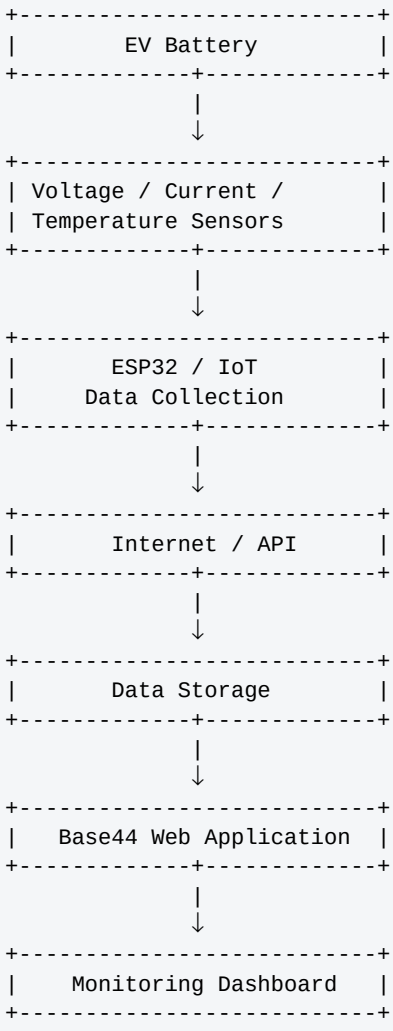
The system is designed so that it can be extended with hardware in the future.

- ESP32
- Arduino
- Voltage sensor
- Current sensor
- Temperature sensor
- Battery Management System (BMS)
- Wi-Fi connectivity

### Proposed Communication Flow

```
EV Battery
    ↓
Sensors / BMS
    ↓
ESP32 / Arduino
    ↓
Wi-Fi / Internet
    ↓
Backend / API
    ↓
Database
    ↓
Web Application
    ↓
User Dashboard
```

# 7. System Architecture




# 8. Working Principle

## Step 1: Battery Data Collection

Battery parameters such as voltage, current, temperature, and SOC are obtained from sensors or a Battery Management System.

### Step 2: Data Processing

The collected parameters are processed to determine the battery's operating condition.

### Step 3: Health Evaluation

The system evaluates battery health using parameters such as SOH, temperature, voltage, current, and charging cycles.

### Step 4: Alert Generation

If a parameter exceeds a predefined limit, the system generates a warning or critical alert.

### Step 5: Web Dashboard

The processed information is displayed through the web application.

### Step 6: Historical Analysis

Battery information can be represented using charts and historical records to understand battery performance.

### Step 7: Maintenance Recommendation

The system provides basic recommendations when battery parameters indicate possible degradation or abnormal operation.

## 9. Technologies Used

### Software

- Base44 AI – web application development
- HTML/CSS/JavaScript concepts – web interface foundation
- AI-assisted application generation
- Web browser

Base44 is used as the application development platform to transform natural-language project requirements into a deployable web application.

### Proposed Hardware

- ESP32 / Arduino
- Voltage sensor
- Current sensor
- Temperature sensor
- EV battery / battery pack
- BMS

Hardware integration is considered an extension of the current web-based mini-project.

## 10. Application Modules

```
Smart Battery Health Monitoring System
■
■■■ Dashboard
■■■ Battery Monitoring
■■■ Health Analysis
■■■ Alerts
■■■ Battery History
■■■ Predictive Maintenance
■■■ Sensor Simulation
■■■ Battery Information
■■■ Settings
```

# 11. Example Battery Parameters

| Parameter | Example Value |
|---|---|
| State of Charge (SOC) | 78% |
| State of Health (SOH) | 92% |
| Battery Voltage | 384 V |
| Battery Current | 42 A |
| Temperature | 31 °C |
| Battery Power | 16.1 kW |
| Charging Cycles | 428 |
| Estimated Range | 186 km |

These values are example/demo values and do not represent measurements from a specific EV.

# 12. Safety and Disclaimer

This project is intended primarily for educational and demonstration purposes. The battery health values, alerts, and predictions generated by the application should not be considered a replacement for professional EV battery diagnostics, Battery Management Systems, or safety-certified battery protection equipment. Actual EV battery systems should use appropriate hardware protection and professionally validated BMS solutions.

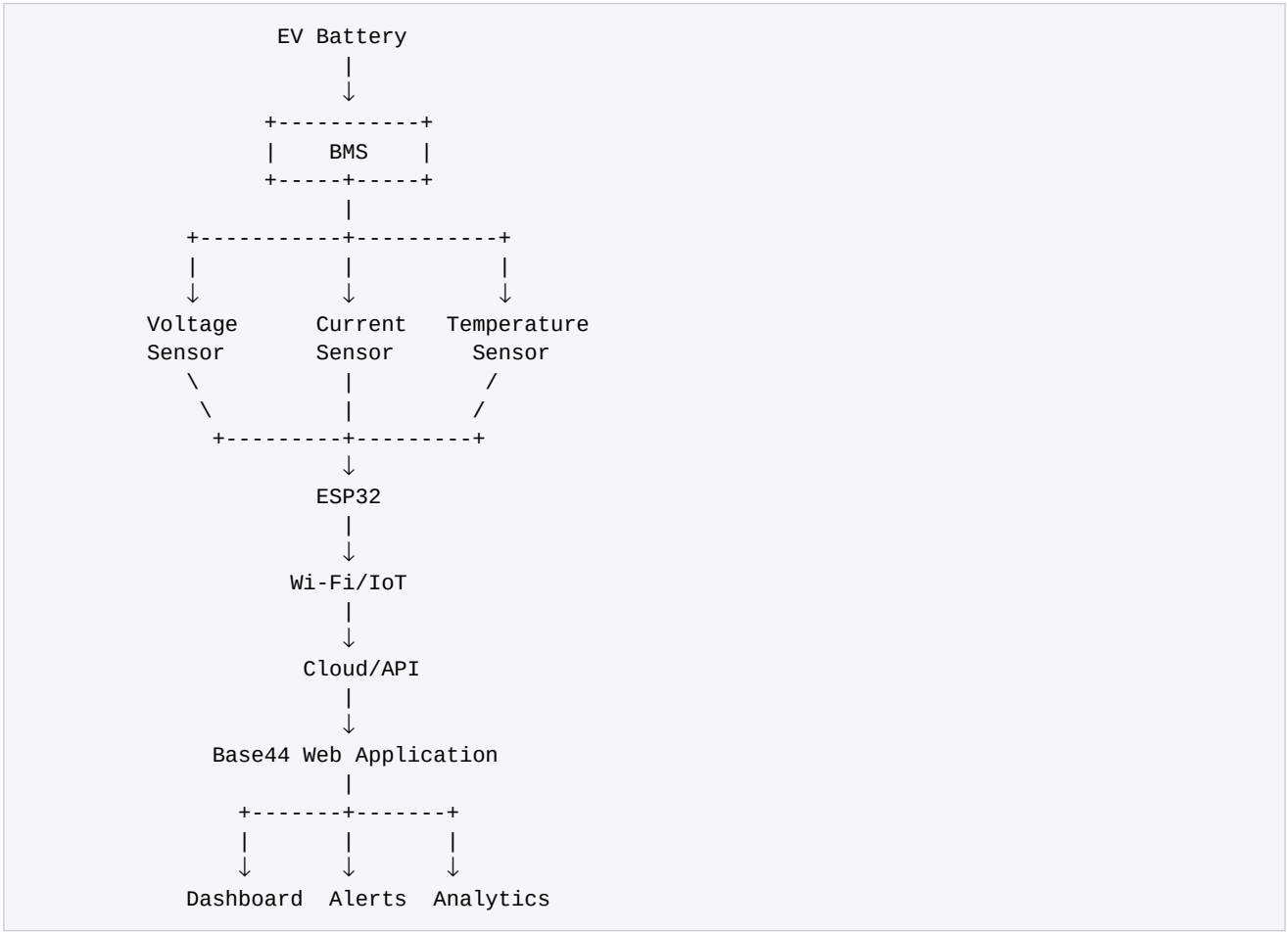
# 13. Advantages

- Easy-to-use web interface
- Centralized battery monitoring
- Visual representation of battery parameters
- Early identification of abnormal conditions
- Historical data analysis
- Maintenance recommendations
- Can be demonstrated without physical hardware using simulation
- Can be extended to IoT hardware
- Suitable for EV-related academic projects
- Demonstrates AI-assisted application development

## 14. Future Scope

- Real-time ESP32 integration
- Real voltage and current sensors
- Temperature sensors
- Direct BMS communication
- Cloud database
- Mobile application
- SMS/email alerts
- Machine-learning-based SOH prediction
- Remaining Useful Life (RUL) prediction
- EV fleet monitoring
- GPS-based vehicle tracking
- Charging station integration
- Battery failure prediction
- Advanced battery degradation analysis

## 15. Future System Architecture




## 16. Testing

| Test Case | Input Condition | Expected Result |
|---|---|---|
| TC01 | Normal temperature | Healthy status |
| TC02 | High temperature | Temperature warning |
| TC03 | Low SOC | Low battery alert |
| TC04 | High voltage | Overvoltage alert |
| TC05 | Low voltage | Undervoltage alert |
| TC06 | Low SOH | Battery health warning |
| TC07 | High current | Overcurrent alert |
| TC08 | Normal battery data | Normal dashboard display |

# 17. Screenshots

Screenshots of the actual Base44 application can be added to the final report and GitHub repository. Suggested screenshots include the Dashboard, Battery Monitoring, Health Analysis, Alerts, Battery History, and Sensor Simulation pages.

## Suggested GitHub image references:

```
![Dashboard](screenshots/dashboard.png)
![Battery Monitoring](screenshots/battery-monitoring.png)
![Alerts](screenshots/alerts.png)
```

# 18. Suggested GitHub Repository Structure

```
Smart-Battery-Health-Monitoring-EV/
■
■■■ README.md
■■■ screenshots/
■    ■■■ dashboard.png
■    ■■■ battery-monitoring.png
■    ■■■ health-analysis.png
■    ■■■ alerts.png
■■■ documentation/
■    ■■■ mini-project-report.pdf
■■■ presentation/
    ■■■ project-presentation.pptx
```

# 19. Academic Significance

- Electronics and Communication Engineering
- Electric Vehicles
- Battery Management Systems
- Sensors
- Embedded Systems
- IoT
- Web Technologies
- Artificial Intelligence
- Data Visualization

The project demonstrates how AI-assisted application development can be used to create a functional monitoring interface for an ECE-oriented engineering problem.

## 20. Conclusion

The Smart Battery Health Monitoring System for EVs provides a web-based platform for monitoring important electric vehicle battery parameters and understanding battery condition. The system demonstrates battery monitoring, health analysis, alerts, historical visualization, sensor simulation, and predictive maintenance concepts.

Using Base44 AI, the web application can be developed from natural-language requirements and deployed as a shareable web application, making it suitable for rapid academic prototyping. The project can be further enhanced by integrating real sensors, ESP32, BMS communication, cloud services, and machine-learning models for more accurate battery health and degradation prediction.

## 21. Project Links

Live Web Application: https://arboreal-volt-pulse-track.base44.app

GitHub Repository: Add your GitHub repository link here.

## 22. References


- Base44 AI – AI-assisted application development platform.
- Electric Vehicle Battery Management System concepts.
- Battery State of Charge (SOC) and State of Health (SOH) concepts.
- IoT-based battery monitoring concepts.
- ESP32 and sensor-based monitoring systems.

# Live Project Link

Smart Battery Health Monitoring System for EVs

https://arboreal-volt-pulse-track.base44.app

Click the link above to open the Base44 web application.